SIAM J. NUMER. ANAL.
Vol. 35, No. 5, pp. 2035–2054, October 1998
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A TWO-LEVEL METHOD WITH BACKTRACKING FOR THE NAVIER–STOKES EQUATIONS*

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Abstract. We consider a two-level method for resolving the nonlinearity in finite element approximations of the equilibrium Navier–Stokes equations. The method yields L^2 and H^1 optimal velocity approximations and an L^2 optimal pressure approximation. The two-level method involves solving one small, nonlinear coarse mesh system, one Oseen problem (hence, linear with positive definite symmetric part) on the fine mesh, and one linear correction problem on the coarse mesh. The algorithm we study produces an approximate solution with the optimal, asymptotic in h, accuracy for any fixed Reynolds number. We do not consider the behavior of the error for fixed h as $Re \to \infty$, i.e., for flows in transition to turbulence.

Key words. Navier–Stokes equations, finite element method, multigrid

AMS subject classifications. 65N30, 76M10

PII. S003614299630230X

1. Introduction.

In this report we consider a two-level method for the resolution of the nonlinear system arising from finite element discretizations of the equilibrium, incompressible Navier–Stokes equations:

$$\begin{cases} -Re^{-1}\Delta\mathbf{u} + \mathbf{u}\cdot\nabla\mathbf{u} + \nabla p = \mathbf{f}, \text{ in } \Omega, \\ \nabla\cdot\mathbf{u} = 0, \text{ in } \Omega, \\ \mathbf{u} = \mathbf{0} \text{ on } \partial\Omega \text{ and } \int_\Omega p\, dx = 0. \end{cases} \tag{1.1}$$

If (1.1) is discretized by the usual Galerkin finite element method, a large system of nonlinear algebraic equations results has the (consistently ordered) block form:

$$\begin{bmatrix} A + N(\mathbf{u}^h) & C \\ C^T & 0 \end{bmatrix} \begin{bmatrix} \mathbf{u}^h \\ p^h \end{bmatrix} = \begin{bmatrix} f^h \\ 0 \end{bmatrix}. \tag{1.2}$$

We are especially interested in efficient methods which can be used for all Reynolds numbers and arbitrary Ladyzhenskaya–Babuska–Brezzi (LBB) stable finite element pairs approximating the velocity $\mathbf{u}$ and pressure p, respectively, of order $k, k \geq 1$. We study herein a two-level method convergent for all Reynolds numbers and all LBB-stable velocity-pressure finite element spaces. A common choice for the solution of (1.2) is linearization by Newton's method, either on a fixed mesh [9, 11, 12] or successive meshes [18, 23, 32], and followed by solving in each Newton step the new linearized system by a multigrid method [31]. Unfortunately, the linearized problems arising from this multilevel Newton method have, at higher Reynolds number, $(1,1)$ blocks $A + N'(\mathbf{u}^n)$ which are both highly nonsymmetric and indefinite. This combination causes difficulties for linear multilevel solvers (and others as well). For example, Turek [36, section 3.1, p. 992] reports that the use of the full linearization can cause instabilities due to this combination of nonsymmetry (the $\mathbf{u}^n \cdot \nabla\mathbf{u}^{n+1}$

*Received by the editors April 19, 1996; accepted for publication (in revised form) October 7, 1997; published electronically August 31, 1998.
http://www.siam.org/journals/sinum/35-5/30230.html
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This author was partially supported by the German Research Association (DFG) under grant TO143/3-1.

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"convection" term) and indefiniteness (the $\mathbf{u}^{n+1} \cdot \nabla \mathbf{u}^n$ "reaction" term), and suggests solving instead an Oseen problem for the update (dropping thereby the $\mathbf{u}^{n+1} \cdot \nabla \mathbf{u}^n$ term). We note that in [36] the nonlinear terms arise from low-order finite elements, an upwind discretization of the convective term and an implicit time discretization of an evolutionary problem. The first two ingredients lead to matrices with an M-matrix structure proven, for example, in Schieweck and Tobiska [29] for the nonconforming Crouzeix–Raviart P1/P0 element. The second ingredient, in particular for small time steps, makes the $(1,1)$ block of (1.2) no longer indefinite. When larger time steps are used to advance in time to a steady state solution, convergence of multilevel methods can slow, as noted in [36] and also in Liu, Liu, and McCormick [23, p. 294] for FAS methods. In the case of the steady Navier–Stokes equations (1.1), the corresponding discrete linearized systems arising from a Newton linearization are more difficult to solve, since there is neither additional stabilization from the time discretization nor M-matrix structure from the combination of low-order elements and upwinding. Because of these difficulties, an Oseen-type linearization has been used extensively in, e.g., [6, 7, 29, 30, 31, 36] to resolve the nonlinearity in (1.2).

As noted in the works cited above, the efficiency of multilevel methods for linearizations of (1.2) depends strongly upon the method of linearization used. We, therefore, consider a method involving an Oseen-type linearization of (1.2) on the finest mesh. Thereby, the $(1,1)$ block of the linearized problem, $A + \tilde{N}'(\mathbf{u}^H)$, will have a positive definite symmetric part. Because of this structure the linear system arising from the Oseen linearization is amenable to solution by multilevel methods as well as other techniques, e.g., [20, 21, 26, 28].

Our method proceeds as follows:

I. Begin by solving the nonlinear system (1.2) for $(\mathbf{u}^H, p^H)$ upon a coarse mesh of width H.
II. Pass (in one step) to the finest mesh of meshwidth h and solve a discrete Oseen problem (with convection term $\mathbf{u}^H \cdot \nabla \mathbf{u}^h$) for the update.
III. Return to the coarsest mesh and solve a linear correction problem (correcting for the omitted "reaction term" $(\mathbf{u}^h \cdot \nabla \mathbf{u}^H)$) and update the fine mesh approximation.

Our algorithm involves no further fixed point iteration. We prove that the approximation produced by I, II, and III is asymptotically as accurate as the approximation produced by solving the fine mesh nonlinear system (1.2) exactly.

In comparison to an Oseen fixed point iteration (with a multilevel solver for the linear system), used in [6, 7, 29, 30, 31, 36], our method is valid for all Reynolds numbers, whereas the fixed point iteration is proven to converge only for small data. Further, our method solves only one discrete Oseen problem, rather than a sequence of fine mesh Oseen problems. In comparison to a multilevel Newton method (see, e.g., Shaidurov [32, Chapter 6.6] or [18]) our method yields, at higher Reynolds numbers, a linear system with $(1,1)$ block having a positive definite symmetric part, rather than being both highly nonsymmetric and indefinite. Further, since we solve only one fixed linear system on the fine mesh, all the intermediate systems required by a linear multilevel solver can be computed without reassembly, whereas a traditional multilevel Newton method, [13, 32] requires reassembly of the linearized problem each time the point of linearization changes, as well as possibly more than one iteration on each level.

To present the two-level method, first consider the usual finite element approximation to (1.1), which can be described as follows. Let $\Pi^h(\Omega)$ denote a division of Ω into d-simplicies, either a priori or self-adaptively [24, 25, 38], where "h" denotes a representative triangle diameter. Conforming velocity-pressure finite element spaces

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$(\mathbf{X}^h, M^h)$ are constructed based upon the mesh $\Pi^h(\Omega)$. Next, the usual finite element approximation $(\hat{\mathbf{u}}, \hat{p})$ is calculated by solving (exactly) the nonlinear system: find $(\hat{\mathbf{u}}, \hat{p}) \in (\mathbf{X}^h, M^h)$ such that, for all $(\mathbf{v}, q) \in (\mathbf{X}^h, M^h)$,

$$a(\hat{\mathbf{u}}, \mathbf{v}) + b(\hat{\mathbf{u}}, \hat{\mathbf{u}}, \mathbf{v}) + (q, \nabla \cdot \hat{\mathbf{u}}) - (\hat{p}, \nabla \cdot \mathbf{v}) = (\mathbf{f}, \mathbf{v}), \tag{1.3}$$

where $(\cdot, \cdot)$ is the ordinary $L^2(\Omega)$ or $(L^2(\Omega))^d$ inner product, $a(\mathbf{u}, \mathbf{v}) := \nu \int_\Omega \nabla \mathbf{u} : \nabla \mathbf{v}\, dx$, and $b(\cdot, \cdot, \cdot)$ is given by either the convective, skew-symmetric or conservation formulation, $b_{1,2,3}(\cdot, \cdot, \cdot)$, respectively, of the $\mathbf{u} \cdot \boldsymbol{\nabla} \mathbf{u}$ term in (1.1). These are given by, respectively, $b_1(\mathbf{u}, \mathbf{v}, \mathbf{w}) := \int_\Omega \mathbf{u} \cdot \boldsymbol{\nabla} \mathbf{v} \cdot \mathbf{w}\, dx$, $b_2(\mathbf{u}, \mathbf{v}, \mathbf{w}) := \frac{1}{2} \int_\Omega [\mathbf{u} \cdot \nabla \mathbf{v} \cdot \mathbf{w} - \mathbf{u} \cdot \nabla \mathbf{w} \cdot \mathbf{v}]\, dx$, and $b_3(\mathbf{u}, \mathbf{v}, \mathbf{w}) := -\int_\Omega \mathbf{u} \cdot \nabla \mathbf{w} \cdot \mathbf{v} dx$. (If a statement is true for all three formulations of the convective term, we omit the index.)

The two-level algorithm we study is as follows. Choose a coarse mesh width H and a fine mesh width h with $H >> h > 0$, and construct associated conforming finite element spaces $(\mathbf{X}^H, M^H)$ and $(\mathbf{X}^h, M^h)$. Now find $(\mathbf{u}^*, p^*)$ as follows:

Algorithm 1.1: Two-level algorithm, fine mesh Oseen solve with coarse mesh correction

Step I: Solve nonlinear problem on coarse mesh

Find $(\mathbf{u}^H, p^H) \in (\mathbf{X}^H, M^H)$ such that, for all $(\mathbf{v}, q) \in (\mathbf{X}^H, M^H)$

$$a(\mathbf{u}^H, \mathbf{v}) + b(\mathbf{u}^H, \mathbf{u}^H, \mathbf{v}) + (q, \nabla \cdot \mathbf{u}^H) - (p^H, \nabla \cdot \mathbf{v}) = (\mathbf{f}, \mathbf{v}).$$

Step II: Update on fine mesh with linear Oseen problem

Find $(\mathbf{u}^h, p^h) \in (\mathbf{X}^h, M^h)$ such that, for all $(\mathbf{v}, q) \in (\mathbf{X}^h, M^h)$

$$a(\mathbf{u}^h, \mathbf{v}) + b(\mathbf{u}^H, \mathbf{u}^h, \mathbf{v}) + (q, \nabla \cdot \mathbf{u}^h) - (p^h, \nabla \cdot \mathbf{v}) = (\mathbf{f}, \mathbf{v}).$$

Step III: Correct on coarse mesh with full linearization

Find $(\mathbf{e}^H, \epsilon^H) \in (\mathbf{X}^H, M^H)$ such that, for all $(\mathbf{v}, q) \in (\mathbf{X}^H, M^H)$

$$\begin{aligned} a(\mathbf{e}^H, \mathbf{v}) + b(\mathbf{u}^H, \mathbf{e}^H, \mathbf{v}) + b(\mathbf{e}^H, \mathbf{u}^H, \mathbf{v}) + (q, \nabla \cdot \mathbf{e}^H) \\ - (\epsilon^H, \nabla \cdot \mathbf{v}) = b(\mathbf{u}^H - \mathbf{u}^h, \mathbf{u}^H, \mathbf{v}), \end{aligned}$$

and set $\mathbf{u}^* = \mathbf{u}^h + \mathbf{e}^H, p^* = p^h + \epsilon^H$.

Our analysis shows that for $(\mathbf{u}^*, p^*)$ to attain the same asymptotic accuracy as $(\hat{\mathbf{u}}, \hat{p})$, the exact solution to the fine mesh equations (1.3), two conditions should be fulfilled. First, the coarse mesh should be fine enough that the error in the coarse mesh approximation is $O(H^k)$ in the energy norm, and $O(H^{k+1})$ in the L^2 norm of the velocities. Second, the step between the coarse and fine meshes should not be too big. In Table 1.1 we present a representative result of the analysis, giving, for the (most commonly used) convective form of the nonlinearity, the largest scaling between coarse and fine meshes for which the desired fine mesh asymptotic accuracy is attained. The table assumes $(\mathbf{X}^\mu, M^\mu), \mu = H, h$, and contains all piecewise polynomials of degree $(k, k-1)$ on $\Pi^\mu(\Omega)$. In the $3d$ case, $\Omega \subset \mathbb{R}^3$, the only change in Table 1.1 that occurs is that "$h \sim H^{3-\epsilon}$" is replaced by $h \sim H^{5/2}$.

Comparing the first and second lines of Table 1.1 shows the important contribution made by the third step in Algorithm 1.1. Step III in Algorithm 1.1 improves the scaling for errors in the energy norm, e.g., from $h \sim H^2$ to $h \sim H^{3-\epsilon}$, if $k = 1$. More importantly, Step III ensures optimal, $O(h^{k+1}), L^2$-velocity errors, still with a superlinear scaling between h and H. As a result, Algorithm 1.1 can be recursively applied either to compute accurate approximate solutions on still finer meshes or to solve the nonlinear coarse mesh problem in Step I.

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TABLE 1.1
Theoretical predictions of scalings sufficient for asymptotically optimal errors in 2-dimensions.

Approximation	Energy norm error $O(h^k)$ provided	L^2 velocity error $O(h^{k+1})$ provided
$(\mathbf{u}^h, p^h)$: Fine mesh Oseen update	$h \sim H^{1+\frac{1}{k}}$	$h \sim H$
$(\mathbf{u}^*, p^*)$: Algorithm 1.1 $b = b_1(\cdot,\cdot,\cdot)$	$h \sim H^{3-\epsilon}, k = 1$ $h \sim H^{1+\frac{2}{k}}, k \geq 2$	$h \sim H^{3/2}, k = 1,$ $h \sim H^{1+\frac{1}{k}}, k \geq 2$
$(\mathbf{u}^*, p^*)$: Algorithm 1.1 $b = b_2(\cdot,\cdot,\cdot)$ or $b = b_3(\cdot,\cdot,\cdot)$	$h \sim H^{1+\frac{1}{k}}, k \geq 1.$	$h \sim H^{1+\frac{1}{k+1}}, k \geq 2.$

Assuming a linear solver becomes available which works well for linearization of (1.2) with a highly nonsymmetric and indefinite $(1,1)$ block $A + N'(\mathbf{u}^H)$, the full linearization can be used on the fine mesh followed by a coarse mesh correction. In this case, the intended purpose of the coarse mesh correction would be to reduce the quadratic error in the fine mesh Newton update. The resulting algorithm would then read as follows:

Algorithm 1.2: Fine mesh Newton update with coarse mesh correction
Step I: Solve nonlinear system on coarse mesh
Find $(\mathbf{u}^H, p^H) \in (\mathbf{X}^H, M^H)$ such that for all $(\mathbf{v}, q) \in (\mathbf{X}^H, M^H)$

$$a(\mathbf{u}^H, \mathbf{v}) + b(\mathbf{u}^H, \mathbf{u}^H, \mathbf{v}) + (q, \nabla \cdot \mathbf{u}^H) - (p^H, \nabla \cdot \mathbf{v}) = (\mathbf{f}, \mathbf{v}).$$

Step II: Update on fine mesh with one Newton step
Find $(\mathbf{u}^h, p^h) \in (\mathbf{X}^h, M^h)$ such that for all $(\mathbf{v}, q) \in (\mathbf{X}^h, M^h)$

$$\begin{aligned} a(\mathbf{u}^h, \mathbf{v}) + b(\mathbf{u}^H, \mathbf{u}^h, \mathbf{v}) + b(\mathbf{u}^h, \mathbf{u}^H, \mathbf{v}) + (q, \nabla \cdot \mathbf{u}^h) \\ - (p^h, \nabla \cdot \mathbf{v}) = (\mathbf{f}, \mathbf{v}) + b(\mathbf{u}^H, \mathbf{u}^H, \mathbf{v}). \end{aligned}$$

Step III: Correct on coarse mesh for the error in Newton step
Find $(\mathbf{e}^H, \epsilon^H) \in (\mathbf{X}^H, M^H)$ such that for all $(\mathbf{v}, q) \in (\mathbf{X}^H, M^H)$

$$\begin{aligned} a(\mathbf{e}^H, \mathbf{v}) + b(\mathbf{u}^H, \mathbf{e}^H, \mathbf{v}) + b(\mathbf{e}^H, \mathbf{u}^H, \mathbf{v}) + (q, \nabla \cdot \mathbf{e}^H) \\ - (\epsilon^H, \nabla \cdot \mathbf{v}) = b(\mathbf{u}^H - \mathbf{u}^h, \mathbf{u}^h - \mathbf{u}^H, \mathbf{v}) \end{aligned}$$

and set $\mathbf{u}^* = \mathbf{u}^h + \mathbf{e}^H$, $p^* = p^h + \epsilon^H$.
The benefits of the extra coarse mesh correction in Algorithm 1.2 seem to be marginal when using the full linearization on the finest mesh (see Table 3.1). The technical reason for this (compared with the greater improvement attained in the work of Xu [40, 41] on elliptic problems) is the lack of exploitable L^∞ convergence bounds for finite element approximations to the Navier–Stokes equations. Thus, additional coarse mesh correction seems to be primarily of interest when solving the Navier–Stokes equation when using a partial linearization on the fine mesh.

Two-level discretization methods seem to go back at least to the early work of Southwell [33] on "group or block relaxation" as well as to projective and multilevel Newton methods (see Witsch [39], Bank [3], Shaidurov [32], and Vainikko [37]). Convergence results for multilevel Newton methods for the Navier–Stokes equations are

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given for small Reynolds numbers in [16] and in Shaidurov [32], and in the nonuniqueness case in [17, 18, 19, 22]. The use of a coarse mesh correction was recently proposed by Xu [40, 41] for semilinear elliptic problems. This additional coarse mesh correction places the resulting two-level method outside the sphere of nonlinear multigrid methods (studied in, e.g., [3, 13, 14, 23, 27, 32]) and projective Newton methods, [37, 39, 2].

Notation: In this report, $W^{k,p} = W^{k,p}(\Omega)$ will denote the usual Sobolev spaces of function with kth distributional derivatives in $L^p(\Omega)$ and equipped with norms $\|\cdot\|_{k,p}$ and seminorms $|\cdot|_{k,p}$, Adams [1]. $\overset{\circ}{W}{}^{1,2}(\Omega)$ is the subspace of $W^{1,2}(\Omega)$ of functions whose trace is zero on $\partial\Omega$ and $L^2_0(\Omega)$ is the space of $L^2(\Omega)$ functions whose mean value is zero. The subscript "p" is suppressed when $p = 2$. The function spaces $\mathbf{X}$ will denote $(\overset{\circ}{W}{}^{1,2}(\Omega))^d$ and M will denote $L^2_0(\Omega) := \{q \in L^2(\Omega) : (q, 1) = 0\}$.

The finite element spaces $\mathbf{X}^h, M^h$ can be thought of as generated from $\mathbf{X}^H, M^H$ by a mesh refinement process, e.g., [24], and therefore nested. It is neither algorithmically necessary, nor needed for the results of our convergence theorems to hold. However, we shall assume them nested since it will simplify our analysis substantially, i.e., $(\mathbf{X}^H, M^H) \subset (\mathbf{X}^h, M^h)$. Further, we assume $(\mathbf{X}^\mu, M^\mu), \mu = h, H$, satisfies the usual inf-sup or LBB stability condition for compatibility of the velocity-pressure spaces:

$$\inf\sup q \in M^\mu \mathbf{v} \in \mathbf{X}^\mu \frac{(q, \boldsymbol{\nabla}\cdot\mathbf{v})}{\|q\|_0\ \|\nabla\mathbf{v}\|_0} \geq \beta > 0,\ \ \mu = H \text{ and } h. \tag{1.4}$$

Moreover, we assume the spaces $(\mathbf{X}^\mu, M^\mu)(\mu = h$ and $H)$ satisfy the approximation properties typical of piecewise polynomials of degree $(k, k-1)$. Specifically, for all $\mathbf{u} \in (\overset{\circ}{W}{}^{1,2}(\Omega) \cap W^{k+1,2}(\Omega))^d, p \in L^2_0(\Omega) \cap W^{k,2}(\Omega)$,

$$\inf_{\substack{\mathbf{v}^h\in\mathbf{X}^h\\ q^h\in M^h}} \{h|\mathbf{u}-\mathbf{v}^h|_1 + \|\mathbf{u}-\mathbf{v}^h\|_0 + h\|p-q^h\|_0\} \leq Ch^{k+1}(|\mathbf{u}|_{k+1} + |p|_k), \tag{1.5}$$

and similarly for $(\mathbf{X}^H, M^H)$.

For $\mu = H, h, \mathbf{V}$ and $\mathbf{V}^\mu$ will denote the space of vector functions in $\mathbf{X}$ and $\mathbf{X}^\mu$, respectively, which are divergence-free and discretely divergence-free, respectively,

$$\mathbf{V} := \{\mathbf{v} \in \mathbf{X} : (q, \nabla\cdot\mathbf{v}) = 0,\ \text{for all } q \in M\},$$
$$\mathbf{V}^\mu := \{\mathbf{v} \in \mathbf{X}^\mu : (q, \nabla\cdot\mathbf{v}) = 0,\ \text{for all } q \in M^\mu\}.$$

It is known [9, 11, 12] that provided the inf-sup condition (1.4) holds, $\mathbf{V}^\mu$ is a well-defined and nontrivial subspace of $\mathbf{X}^\mu$, and that a function in $\mathbf{V}$ can be approximated in $\mathbf{V}^\mu$ with the same asymptotic accuracy as in $\mathbf{X}^\mu$.

Since we study in section 3 the scalings between the coarse and fine meshes that suffice for optimal, $O(h^{k+1})$, L^2-velocity errors, we assume throughout that the nonsingular solution $(\mathbf{u}, p)$ of (1.1) we approximate is such that the linearized adjoint problem is "$W^{2,2}$ regular" in the usual sense [10, 11] and which we precisely formulate in section 3.

2. The error in the energy norm. This section gives the basic error analysis of Algorithm 1.1 in the natural energy norm for the Navier–Stokes equations, given by

$$|||(\mathbf{v}, q)||| := (|\mathbf{v}|_1^2 + \|q\|_0^2)^{1/2}.$$

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This error analysis proceeds in the product space $(\mathbf{X}^h, M^h)$ rather than following the classical approach of first estimating the component of the error in $\mathbf{V}^h$ since the approximation $\mathbf{u}^*$ does *not* belong to the discretely divergence-free space $\mathbf{V}^h$ due to the coarse mesh correction. In section 3, error estimates are proven which yield scalings sufficient as well for L^2-optimality of the approximate fluid velocities.

We shall also state without proof the results for Algorithm 1.2 as the methods of proof are quite analogous. The choice of the formulation of the nonlinear term (convective, skew-symmetric, or conservation) alters the theoretical prediction, with the convective form giving somewhat better predicted results than the other two.

THEOREM 2.1. *Suppose* $\mathbf{X}^H \subset \mathbf{X}^h \subset \mathbf{X}, M^H \subset M^h \subset M$, *and each pair* $(\mathbf{X}^H, M^H), (\mathbf{X}^h, M^h)$ *satisfies the LBB condition,* (1.4), *and the approximation assumption,* (1.5). *Let* $(\mathbf{u}, p)$ *be a nonsingular solution of the Navier–Stokes equations that is smooth enough,* $\mathbf{u} \in (\overset{\circ}{W}{}^{1,2} \cap W^{k+1,2} \cap W^{1,\infty})^d$ *and* $p \in (L_0^2 \cap W^{k,2})$. *Suppose the linearized dual problem to* (1.1) *is* $W^{2,2}$ *regular,* $0 < h < H$ *and* H *is sufficiently small. Let* $s = \frac{1}{2}$ *if* $d = 3$ *and* $s = \epsilon > 0$ *if* $d = 2$. *Then, for the convective nonlinearity,* $b := b_1(\cdot,\cdot,\cdot)$, *there is a constant* $C = C(s, Re, \mathbf{u}, p)$ *such that*

$$|||(\mathbf{u} - \mathbf{u}^*, p - p^*)||| \le C(h^k + H^{k+2} + H^{2k+1-s}),$$
$$|||(\mathbf{u} - \mathbf{u}^*, p - p^*)||| \le C(\mathbf{u}, p)h^{k+1}, \text{ provided } h = O(H^{\min\{1+2/k, 2+\frac{1-s}{k}\}}).$$

If the skew-symmetrized form $b = b_2(\cdot,\cdot,\cdot)$ *or the conservation form* $b = b_3(\cdot,\cdot,\cdot)$ *is chosen instead, then*

$$|||(\mathbf{u} - \mathbf{u}^*, p - p^*)||| \le C(h^k + H^{k+1} + H^{2k+1-s}),$$
$$|||(\mathbf{u} - \mathbf{u}^*, p - p^*)||| \le C(\mathbf{u}, p)h^{k+1}, \quad \text{provided } h = O(H^{1+1/k}).$$

REMARK 2.1 (on the assumed regularity of $\mathbf{u}$). *The assumption that the true solution* $\mathbf{u} \in (W^{1,\infty})^d$ *simplifies the analysis at several points. If* $\mathbf{u} \notin (W^{1,\infty})^d$, *then by adding and subtracting terms at several places and judiciously using discrete imbedding inequalities, we arrive at essentially the same error bounds as in Theorem* 2.1, *with an extra* $|ln(H)|^{1-1/d}$ *factor multiplying the powers of "H" on the right-hand side.*

The assumption of $W^{2,2}$ *regularity for the linearized dual problem can be demonstrated, e.g., planar polygonal domains by using* $W^{2,2}$ *regularity of the Stokes problem for these domains and a bootstrap argument. If* $W^{2,2}$ *regularity fails for the Stokes problem, we expect it also to fail for the linearized dual problem. In this case, we expect at least* $W^{1+\alpha}$ *regularity for some* $\alpha > 0$. *The "target"* L^2 *accuracy is then* $O(h^{1+\alpha})$. *The analysis herein can be repeated for the* $W^{1+\alpha}$ *regularity case, showing (appropriately modified) improvement due to Step* III.

The remainder of this section will consist of the proof of Theorem 2.1. First note that Algorithm 1.1 can be reformulated in the product spaces $\mathbf{Y}^H := (\mathbf{X}^H, M^H)$, $\mathbf{Y}^h := (\mathbf{X}^h, M^h)$, and $\mathbf{Y} := (\mathbf{X}, M)$. Indeed, introducing the continuous bilinear forms $A^H : \mathbf{Y} \times \mathbf{Y} \to \mathbb{R}$ and $B^H : \mathbf{Y} \times \mathbf{Y} \to \mathbb{R}$ given by

$$A^H[(\mathbf{u}, p); (\mathbf{v}, q)] := a(\mathbf{u}, \mathbf{v}) + b(\mathbf{u}^H, \mathbf{u}, \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) + (q, \nabla \cdot \mathbf{u}),$$

$$B^H[(\mathbf{u}, p); (\mathbf{v}, q)] := a(\mathbf{u}, \mathbf{v}) + b(\mathbf{u}^H, \mathbf{u}, \mathbf{v}) + b(\mathbf{u}, \mathbf{u}^H, \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) + (q, \nabla \cdot \mathbf{u}),$$

we can reformulate Step II and Step III of Algorithm 1.1 in the product space to be:

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Step II: $A^H[(\mathbf{u}^h, p^h); (\mathbf{v}, q)] = (\mathbf{f}, \mathbf{v})$, for all $(\mathbf{v}, q) \in \mathbf{Y}^h$,

Step III: $B^H[(\mathbf{e}^H, \epsilon^H); (\mathbf{v}, q)] = b(\mathbf{u}^H - \mathbf{u}^h, \mathbf{u}^H, \mathbf{v})$, for all $(\mathbf{v}, q) \in \mathbf{Y}^H$.

The bilinear forms $A^H[\cdot,\cdot]$ and $B^H[\cdot,\cdot]$ are easily shown to be continuous on the product space $\mathbf{Y} \times \mathbf{Y}$ with continuity constant depending on $|\mathbf{u}^H|_1$. Thus, provided $|\mathbf{u}^H|_1$ is bounded uniformly in H, the bilinear forms $A^H[\cdot,\cdot]$ and $B^H[\cdot,\cdot]$ are continuous on $\mathbf{Y} \times \mathbf{Y}$ uniformly in H. Uniform boundedness in H of $|\mathbf{u}^H|_1$ follows directly from Step I (setting $\mathbf{v} = \mathbf{u}^H$ and $q = p^H$) when the skew-symmetrized trilinear form of b is used. If the convective or conservation form of b is used, than boundedness of $|\mathbf{u}^H|_1$ follows by the assumption that H is small enough, implying that $\mathbf{u}^H$ is close to $\mathbf{u}$:

$$|\mathbf{u}^H|_1 \le |\mathbf{u}^H - \mathbf{u}|_1 + |\mathbf{u}|_1 \le CH(|\mathbf{u}|_2 + |p|_1) + |\mathbf{u}|_1 \le C(\mathbf{u}, p).$$

Thus, in either case, $A^H[\cdot;\cdot]$ and $B^H[\cdot;\cdot]$ are continuous on $\mathbf{Y} \times \mathbf{Y}$ uniformly in H.

In order to guarantee the solvability of the problems in Steps II and III, respectively, we give Lemma 2.1, which is known from the theory of mixed methods (see, e.g., Brezzi and Fortin [4]). We will formulate it in the space $\mathbf{Y} = \mathbf{X} \times M$; however, it is also applicable to the spaces $\mathbf{Y}^\mu \times M^\mu$ with $\mu = h$ or $\mu = H$. With a given continuous bilinear form $c : \mathbf{X} \times \mathbf{X} \to \mathbb{R}$ and given continuous linear forms $\langle \mathbf{f}, \cdot \rangle : \mathbf{X} \to \mathbb{R}$, $\langle g, \cdot \rangle : M \to \mathbb{R}$, we consider the generalized Stokes problem

$$\begin{aligned} &\text{Find } (\mathbf{u}, p) \in \mathbf{X} \times M \text{ such that for all } (\mathbf{v}, q) \in \mathbf{X} \times M, \\ &c(\mathbf{u}, \mathbf{v}) - (p,\ \nabla \cdot \mathbf{v}) + (q,\ \nabla \cdot \mathbf{u}) = \langle \mathbf{f}, \mathbf{v} \rangle + \langle g, q \rangle. \end{aligned} \tag{2.1}$$

LEMMA 2.1. *Suppose that*

$$\inf_{q \in M} \sup_{\mathbf{v} \in \mathbf{X}} \frac{(q, \nabla \cdot \mathbf{v})}{\|q\|_0 |\mathbf{v}|_1} \ge \beta > 0 \tag{2.2}$$

and that the bilinear form c satisfies

$$\inf_{\mathbf{u} \in \mathbf{V}} \sup_{\mathbf{v} \in \mathbf{V}} \frac{c(\mathbf{u}, \mathbf{v})}{|\mathbf{u}|_1 |\mathbf{v}|_1} \ge \alpha_0 > 0, \tag{2.3}$$

$$\inf_{\mathbf{v} \in \mathbf{V}} \sup_{\mathbf{u} \in \mathbf{V}} \frac{c(\mathbf{u}, \mathbf{v})}{|\mathbf{u}|_1 |\mathbf{v}|_1} \ge \alpha_0 > 0. \tag{2.4}$$

Then the continuous bilinear form $F : \mathbf{Y} \times \mathbf{Y} \to \mathbb{R}$ defined by

$$F[(\mathbf{u}, p); (\mathbf{v}, q)] := c(\mathbf{u}, \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) + (q, \nabla \cdot \mathbf{u}) \tag{2.5}$$

satisfies the inf-sup *conditions on the product space $\mathbf{Y} \times \mathbf{Y}$, which means that there is a constant $\gamma = \gamma(\alpha_0, \beta, \|c\|) > 0$ such that*

$$\inf_{(\mathbf{u},p) \in \mathbf{Y}} \sup_{(\mathbf{v},q) \in \mathbf{Y}} \frac{F[(\mathbf{u}, p); (\mathbf{v}, q)]}{|||(\mathbf{u}, p)|||\ |||(\mathbf{v}, q)|||} \ge \gamma, \tag{2.6}$$

$$\inf_{(\mathbf{v},q) \in \mathbf{Y}} \sup_{(\mathbf{u},p) \in \mathbf{Y}} \frac{F[(\mathbf{u}, p); (\mathbf{v}, q)]}{|||(\mathbf{u}, p)|||\ |||(\mathbf{v}, q)|||} \ge \gamma. \tag{2.7}$$

Proof. See Brezzi and Fortin [4, Chapter II, Thm. 1.1] or Girault and Raviart [9, Chapter I, Thm. 4.1]. ∎

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REMARK 2.2. *Note that* (2.6), (2.7) *imply that the mapping* $\boldsymbol{\phi} : \mathbf{Y} \to \mathbf{Y}^*$ *defined by*

$$\langle \boldsymbol{\phi}(\mathbf{u}, p); (\mathbf{v}, q)\rangle := F[(\mathbf{u}, p); (\mathbf{v}, q)], \quad \textit{for all } (\mathbf{u}, p), (\mathbf{v}, q) \in \mathbf{Y}$$

is an isomorphism from $\mathbf{Y}$ *onto* $\mathbf{Y}^*$.

REMARK 2.3. *Applying Lemma* 2.1 *for the finite-dimensional spaces* $\mathbf{Y}^H$ *or* $\mathbf{Y}^h$, *only one of the assumptions* (2.3), (2.4) *is needed, because in finite dimensions uniqueness already implies surjectivity. Moreover, in the case that* c *is symmetric on* $\mathbf{V}$, *the assumptions* (2.3) *and* (2.4) *are equivalent. Finally, note that* $\mathbf{V}$*-ellipticity of* c *is sufficient for* (2.3) *and* (2.4).

Now let us study the application of Lemma 2.1 on the problems to be solved in Steps II and III of Algorithm 1.1. For investigating Step II we set

$$c(\mathbf{u}, \mathbf{v}) := a(\mathbf{u}, \mathbf{v}) + b(\mathbf{u}^H, \mathbf{u}, \mathbf{v})$$

and get coercivity immediately, hence, (2.3) and (2.4), for the skew-symmetrized form $b = b_2$:

$$c(\mathbf{v}, \mathbf{v}) = \nu|\mathbf{v}|_1^2 \quad \forall \mathbf{v} \in \mathbf{X}.$$

In case of the convective form $b = b_1$ and conservation form $b = b_3$, we have

$$c(\mathbf{v}, \mathbf{v}) = \nu|\mathbf{v}|_1^2 - \frac{1}{2}(\nabla \cdot \mathbf{u}^H, \mathbf{v} \cdot \mathbf{v}).$$

Consider the $(\nabla \cdot \mathbf{u}^H, \mathbf{v} \cdot \mathbf{v})$ term. Since $\nabla \cdot \mathbf{u}^H$ is orthogonal to Q^H it follows that for any $q^H \in Q^H$

$$(\nabla \cdot \mathbf{u}^H, \mathbf{v} \cdot \mathbf{v}) = (\nabla \cdot \mathbf{u}^H, \mathbf{v} \cdot \mathbf{v} - q^H).$$

Picking q^H to be the L^2 projection of $\mathbf{v} \cdot \mathbf{v} - \int_\Omega |\mathbf{v}|^2 dx$ into Q^H gives that $\mathbf{v} \cdot \mathbf{v} - \int_\Omega |\mathbf{v}|^2 dx - q^H$ is again orthogonal to Q^H. Thus, for this choice of q^H, for any $\chi^H \in Q^H$

$$\begin{aligned} |(\nabla \cdot \mathbf{u}^H, \mathbf{v} \cdot \mathbf{v})| &= |(\nabla \cdot \mathbf{u}^H - \chi^H, \mathbf{v} \cdot \mathbf{v} - q^H)| \\ &\le |(\nabla \cdot \mathbf{u}^H - x^H, w - q^H)| + |(\nabla \cdot \mathbf{u}^H - \chi^H, \bar{w})|, \end{aligned}$$

where $w = \mathbf{v} \cdot \mathbf{v} - \bar{w}$ and $\bar{w} = \int_\Omega |\mathbf{v}|^2 dx$. Since $\bar{w}$ is constant, $\mathbf{u}^H = 0$ on $\partial\Omega$ and $\int_\Omega \chi^H dx = 0$ it follows that $(\nabla \cdot \mathbf{u}^H - \chi^H, \bar{w}) = 0$. Consider thus the first term $(\nabla \cdot \mathbf{u}^H - \chi^H, w - q^H)$. First note that w has mean value zero so w is approximable by functions q^H in Q^H. Further, by continuity of the multiplication operator $W^{\beta,4} \times W^{\beta,4} \to W^{\beta,2}$ (Grisvard [10, Thm 1.4.4.2, p. 28], for $0 \le \beta \le 1$, $|w|_\beta = |\mathbf{v} \cdot \mathbf{v}|_\beta \le C|\mathbf{v}|^2_{W^{\beta,4}}$. Further, by the Sobolev embedding theorem $|\mathbf{v}|_{W^{\beta,4}} \le C|\mathbf{v}|_{W^{1,2}}$, for $0 \le \beta \le \frac{1}{2}$ if $d = 2$ and for $0 \le \beta \le \frac{1}{4}$ if $d = 3$. Thus, for $w \in L^2_0(\Omega) \cap W^{\beta,2}(\Omega)$, ($\beta = \frac{1}{2}$ if $d = 2, \beta = \frac{1}{4}$ if $d = 3$)

$$\begin{aligned} |(\nabla \cdot \mathbf{u}^H - \chi^H, w - q^H)| &\le ||\nabla \cdot \mathbf{u}^H - \chi^H||_0 ||w - q^H||_0 \\ &\le ||\nabla \cdot \mathbf{u}^H - \chi^H||_0 C H^\beta |w|_\beta \\ &\le C||\nabla \cdot \mathbf{u}^H - \chi^H||_0 \cdot H^\beta |\mathbf{v}|_1^2. \end{aligned}$$

Using this last expression for the inequality for $c(\mathbf{v}, \mathbf{v})$ gives, for all $\mathbf{v} \in \mathbf{X}$,

$$c(\mathbf{v}, \mathbf{v}) \ge (\nu - C^* H^\beta \inf_{\chi^H \in Q^H} ||\nabla \cdot \mathbf{u}^H - \chi^H||_0)|\mathbf{v}|_1^2 \ge (\nu - C^* H^\beta ||\nabla \cdot \mathbf{u}^H||_0)|\mathbf{v}|_1^2,$$

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where $\beta = \frac{1}{2}$ in $2d$ and $\frac{1}{4}$ in $3d$ and C^* is independent of ν. Coercivity thus follows (1) if $\nabla \cdot \mathbf{u}^H \in Q^H$ (as occurs with certain higher order choices of $(\mathbf{X}^H, Q^H)$ or if (2) $H^\beta ||\nabla \cdot u^H||_0 (\leq H^\beta ||\nabla(u - u^H)||_0)$ is small enough with respect to ν.

If the usual Galerkin method is used for the coarse mesh discretization, we believe a smallness condition on H without regard to ν is needed to ensure positive definiteness of $c(\cdot,\cdot)$ when the convective form is used. However, this can be eliminated by altering the coarse mesh variational formulation (see Remark 2.4 smallness condition). Thus, by Lemma 2.1 there is a $\gamma > 0$, independent of h and H, such that

$$\inf \ \sup(\mathbf{u}, p) \in \mathbf{Y}^h(\mathbf{v}, q) \in \mathbf{Y}^h \frac{A^H[(\mathbf{u}, p); (\mathbf{v}, q)]}{|||(\mathbf{u}, p)||| \ |||(\mathbf{v}, q)|||} \geq \gamma > 0. \tag{2.8}$$

Remark 2.4. *Consider Algorithm* 1.1. *If the convective form of the partially lagged nonlinearity is used in Step* 2 *and the fine mesh linear system arising from Step* 2 *of Algorithm* 1.1 *is singular, Step* 1 *of Algorithm* 1.1 *may be modified by using the skew-symmetrized nonlinearity plus a least-squares term to control* $||\nabla \cdot \mathbf{u}^H||$ *on the coarse mesh, as follows. Replace Step* 1 *by the following: Calculate* $(\mathbf{u}^H, p^H) \in (\mathbf{X}^H, Q^H)$ *satisfying*

$$\begin{aligned} &\delta(\nabla \cdot \mathbf{u}^H \nabla \cdot \mathbf{v}^H) + \nu a(\mathbf{u}^H, \mathbf{v}^H) \frac{1}{2}(\mathbf{u}^H \cdot \nabla \mathbf{u}^H, \mathbf{v}^H) - \frac{1}{2}(\mathbf{u}^H \cdot \nabla \mathbf{v}^H, \mathbf{u}^H) \\ &- (p^H, \nabla \cdot \mathbf{v}^H) + (q^H, \nabla \cdot \mathbf{u}^H) = (\mathbf{f}, \mathbf{v}^H), \ \textit{for all } (\mathbf{v}^H, q^H) \in (\mathbf{X}^H, Q^H), \end{aligned}$$

where δ *is chosen to ensure* $||\nabla \cdot \mathbf{u}^H||_0$ *is sufficiently small. Indeed, setting* $\mathbf{v}^H = \mathbf{u}^H, q^H = p^H$ *in this modified coarse mesh discretization gives*

$$\delta ||\nabla \cdot \mathbf{u}^H||_0^2 + \nu ||\nabla \mathbf{u}^H||_0^2 = (\mathbf{f}, \mathbf{u}^H),$$

so that

$$||\nabla \cdot \mathbf{u}^H|| \leq \delta^{-1} ||f||_*.$$

Picking $\delta \geq 2||f||_* C^* \frac{H^\beta}{\nu}$ *then ensures that, at Step* 2, *for all* $v \in \mathbf{X}^h$,

$$c(\mathbf{v}, \mathbf{v}) \geq (\nu - C^* H^\beta ||\nabla \cdot \mathbf{u}^H||_0)||\mathbf{v}||_1^2 \geq \frac{\nu}{2}|\mathbf{v}|_1^2.$$

The additional term "$\delta(\nabla \cdot \mathbf{u}^H, \nabla \cdot \mathbf{v}^H)$*" is conforming (since* $\nabla \cdot \mathbf{u} = 0$*) and does not materially alter either the subsequent analysis, or the resulting scalings between* h *and* H. *Additive least-squares terms of this type have been used, e.g., in streamline diffusion discretizations of* (1.1), [35, 15].

Considering Step III, set

$$c(\mathbf{u}, \mathbf{v}) := a(\mathbf{u}, \mathbf{v}) + b(\mathbf{u}^H, \mathbf{u}, \mathbf{v}) + b(\mathbf{u}, \mathbf{u}^H, \mathbf{v})$$

and the condition (2.3), (2.4) follows from the assumption that $\mathbf{u}$ is a nonsingular solution of the Navier–Stokes equations and $\mathbf{u}^H$ is close enough to $\mathbf{u}$. In more detail, if $\mathbf{u}$ is a nonsingular solution of (1.1), then [8, 9] there is a constant $\gamma^* = \gamma^*(\mathbf{u})$ such that

$$\inf_{\mathbf{v} \in \mathbf{V}} \sup_{\mathbf{w} \in \mathbf{V}} \frac{a(\mathbf{v}, \mathbf{w}) + b(\mathbf{u}, \mathbf{v}, \mathbf{w}) + b(\mathbf{v}, \mathbf{u}, \mathbf{w})}{|\mathbf{v}|_1 \ |\mathbf{w}|_1} \geq \gamma^* > 0. \tag{2.9}$$

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From (2.9) it simply follows that, for $|\mathbf{u}-\mathbf{u}^H|_1$ small enough (which is the case for all H with $0 < H \le H_0$),

$$\inf_{\mathbf{v}\in\mathbf{V}} \sup_{\mathbf{w}\in\mathbf{V}} \frac{a(\mathbf{v},\mathbf{w}) + b(\mathbf{u}^H,\mathbf{v},\mathbf{w}) + b(\mathbf{v},\mathbf{u}^H,\mathbf{w})}{|\mathbf{v}|_1\ |\mathbf{w}|_1} \ge \frac{\gamma^*}{2} > 0. \tag{2.10}$$

Later we shall apply Lemma 2.1 on the subspaces $\mathbf{Y}^H$ and $\mathbf{Y}^h$. In order to guarantee the assumptions (2.3) and (2.4), respectively, we will need an inf-sup condition like (2.9) but on the spaces $\mathbf{V}^H$ and $\mathbf{V}^h$. That this follows from (2.9) is a basic fact from [8, 9, 11] about the approximation of nonsingular solutions of (1.1). Thus, we have, for B^H,

$$\inf_{(\mathbf{u},p)\in\mathbf{Y}^H} \sup_{(\mathbf{v},q)\in\mathbf{Y}^H} \frac{B^H[(\mathbf{u},p);(\mathbf{v},q)]}{|||(\mathbf{u},p)|||\ |||(\mathbf{v},q)|||} \ge \gamma > 0 \tag{2.11}$$

and

$$\inf_{(\mathbf{v},q)\in\mathbf{Y}^h} \sup_{(\mathbf{u},p)\in\mathbf{Y}^h} \frac{B^H[(\mathbf{u},p);(\mathbf{v},q)]}{|||(\mathbf{u},p)|||\ |||(\mathbf{v},q)|||} \ge \gamma > 0. \tag{2.12}$$

In particular, (2.11) gives the unique solvability of the correction Step III. ∎

Before we study the error of Algorithm 1.1 in detail, we formulate a set of inequalities for estimating the trilinear forms b_1, b_2, and b_3 and sufficient conditions which guarantee the unique solvability of the linear problems in Steps II and III, respectively.

The next lemma requires the use of the Sobolev imbedding theorem in fractional order Sobolev spaces on domains with polyhedral boundaries. Such boundaries satisfy the strong local Lipschitz property (see Adams [1, Chapter IV, 4.5, pp. 66–67]). The spaces $W^{s,p}(\Omega)$ are well defined for noninteger s and domains with Lipschitz boundaries (Adams [1, Chapter VII]). The imbedding theorems we use are noninteger generalizations of the Sobolev imbedding theorems given in Adams [1, Chapter VII, Theorems 7.57, 7.58, pp. 217–219]. Trilinear form bounds of the type given in the next lemma appear, for example, in Témam [34], and Constantin and Foias [5], under conditions on $\partial\Omega$, which Lemma 2.2 shows are unnecessarily restrictive.

LEMMA 2.2. *Suppose the boundary of the domain Ω satisfies the strong Lipschitz condition of Adams [1, p. 66]. Let $b(\cdot,\cdot,\cdot)$ be the convective form $b = b_1$ or the skew-symmetrized form $b = b_2$. Then the following estimates hold:*

(a) $|b_1(\mathbf{u},\mathbf{v},\mathbf{w})| \le C|\mathbf{u}|_1\|\mathbf{v}\|_{1,3}\|\mathbf{w}\|_0$, *for all* $\mathbf{u},\mathbf{w}\in\mathbf{X}, \mathbf{v}\in\mathbf{W}^{1,3}(\Omega)$;
(b) $|b_1(\mathbf{u},\mathbf{v},\mathbf{w})| \le C\|\mathbf{u}\|_0|\mathbf{v}|_{1,\infty}\|\mathbf{w}\|_0$, *for all* $\mathbf{u},\mathbf{w}\in\mathbf{X}, \mathbf{v}\in\mathbf{W}^{1,\infty}(\Omega)$;
(c) $|b_2(\mathbf{u},\mathbf{v},\mathbf{w})| \le C(\|\mathbf{u}\|_0|\mathbf{v}|_{1,\infty} + \|\mathbf{u}|_1\|\mathbf{v}\|_{0,\infty})\|\mathbf{w}\|_0$, *for all* $\mathbf{u},\mathbf{w}\in\mathbf{X}, \mathbf{v}\in \mathbf{W}^{1,\infty}(\Omega)$;
(d) $|b_2(\mathbf{u},\mathbf{v},\mathbf{w})| \le C\|\mathbf{u}\|_0(|\mathbf{v}|_{1,\infty}\|\mathbf{w}\|_0 + \|\mathbf{v}\|_2|\mathbf{w}|_1)$, *for all* $\mathbf{u},\mathbf{w}\in\mathbf{X}, \mathbf{v}\in\mathbf{W}^{2,2}(\Omega)\cap\mathbf{W}^{1,\infty}(\Omega)$;
(e) $|b(\mathbf{u},\mathbf{v},\mathbf{w})| \le C|\mathbf{u}|_1|\mathbf{v}|_1|\mathbf{w}|_1$, *for all* $\mathbf{u},\mathbf{v},\mathbf{w}\in\mathbf{X}$;
(f) $|b(\mathbf{u},\mathbf{v},\mathbf{w})| \le C\|\mathbf{u}\|_0^{1-s}|\mathbf{u}|_1^s|\mathbf{v}|_1|\mathbf{w}|_1$, *for all* $\mathbf{u},\mathbf{v},\mathbf{w}\in\mathbf{X}$;
(g) $|b(\mathbf{u},\mathbf{v},\mathbf{w})| \le C\|\mathbf{u}\|_0|\mathbf{v}|_1\|\mathbf{w}\|_2$, *for all* $\mathbf{u},\mathbf{v}\in\mathbf{X}, \mathbf{w}\in\mathbf{W}^{2,2}(\Omega)$;
(h) $|b(\mathbf{u},\mathbf{v},\mathbf{w})| \le C|\mathbf{u}|_1\|\mathbf{v}\|_0\|\mathbf{w}\|_2$, *for all* $\mathbf{u},\mathbf{v},\in\mathbf{X}, \mathbf{w}\in\mathbf{W}^{2,2}(\Omega)$;
(i) $|b(\mathbf{u},\mathbf{v},\mathbf{w})| \le C|\mathbf{u}|_1\|\mathbf{v}\|_2\|\mathbf{w}\|_0$, *for all* $\mathbf{u},\mathbf{w}\in\mathbf{X}, \mathbf{v}\in\mathbf{W}^{2,2}(\Omega)$,

where $s = \epsilon > 0$ is arbitrarily small for $d = 2$ and $s = \frac{1}{2}$ for $d = 3$. The conservation form $b = b_3(\cdot,\cdot,\cdot)$ satisfies the same inequalities as $b_2(\cdot,\cdot,\cdot)$, (c)–(i) *above.*

Proof. For (a) and (b), note that the Hölder's inequality gives

$$|(\mathbf{u}\cdot\nabla\mathbf{v},\mathbf{w})| \le \|\mathbf{u}\|_{0,p}\|\nabla\mathbf{v}\|_{0,q}\|\mathbf{w}\|_{0,r}, \quad \frac{1}{p}+\frac{1}{q}+\frac{1}{r} = 1.$$

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Setting $p = 6, q = 3, r = 2$ and $p = 2, q = \infty, r = 2$, respectively, and using the imbedding $\overset{\circ}{W}{}^{1,2}(\Omega) \to L^6(\Omega)$ give (a) and (b). To prove (c), we use integration by parts and Hölder's inequality, giving

$$\begin{aligned} b_2(\mathbf{u}, \mathbf{v}, \mathbf{w}) &= (\mathbf{u} \cdot \nabla \mathbf{v}, \mathbf{w}) + \frac{1}{2}(\nabla \cdot \mathbf{u}, \mathbf{v} \cdot \mathbf{w}), \\ |b_2(\mathbf{u}, \mathbf{v}, \mathbf{w}) &\le C(\|\mathbf{u}\|_0 |\mathbf{v}|_{1,\infty} \|\mathbf{w}\|_0 + |\mathbf{u}|_1 \|\mathbf{v}\|_{0,\infty} \|\mathbf{w}\|_0). \end{aligned}$$

Now (d) follows directly using Sobolev's imbedding $W^{2,2}(\Omega) \to L^\infty(\Omega)$. Indeed,

$$\begin{aligned} |b_2(\mathbf{u}, \mathbf{v}, \mathbf{w})| &\le \frac{1}{2}|(\mathbf{u} \cdot \nabla \mathbf{v}, \mathbf{w})| + \frac{1}{2}|(\mathbf{u} \cdot \nabla \mathbf{w}, \mathbf{v})| \\ &\le C\|\mathbf{u}\|_0 |\mathbf{v}|_{1,\infty} \|\mathbf{w}\|_0 + C\|\mathbf{u}\|_0 |\mathbf{w}|_1 \|\mathbf{v}\|_{0,\infty} \\ &\le C\|\mathbf{u}\|_0 (|\mathbf{v}|_{1,\infty} \|\mathbf{w}\|_0 + \|\mathbf{v}\|_2 |\mathbf{w}|_1). \end{aligned}$$

Part (e) is proved by applying Sobolev's imbedding $W^{1,2}(\Omega) \to L^6(\Omega)$ and Hölder's inequality for $p = 6, q = 2, r = 6$. To show (f), we use Sobolev spaces of fractional order. We consider first $b = b_1$. Hölder's inequality implies

$$|(\mathbf{u} \cdot \nabla \mathbf{v}, \mathbf{w})| \le \|\mathbf{u}\|_{0,p} |\mathbf{v}|_1 \|\mathbf{w}\|_{0,r}, \quad \frac{1}{p} + \frac{1}{r} = \frac{1}{2}.$$

Consider the case $d = 2$. Given $\epsilon > 0$, arbitrary, we use the imbedding $W^{\epsilon,2}(\Omega) \to L^p(\Omega)$ with $p = \frac{2}{1-\epsilon}$, which gives

$$\|\mathbf{u}\|_{0,p} \le C(\epsilon)\|\mathbf{u}\|_\epsilon, \text{ for all } \mathbf{u} \in (W^{\epsilon,2}(\Omega))^2.$$

Since $r = \frac{1}{\epsilon}$ and $W^{1,2}(\Omega) \to L^r(\Omega)$ for any $r \in [2, \infty)$, we obtain

$$\|\mathbf{w}\|_{0,r} \le C(\epsilon)|\mathbf{w}|_1, \text{ for all } \mathbf{w} \in \mathbf{X}.$$

For the three-dimensional case, $d = 3$, we use the imbeddings $W^{\frac{1}{2},2}(\Omega) \to L^3(\Omega)$ and $W^{1,2}(\Omega) \to L^6(\Omega)$ and get

$$|(\mathbf{u} \cdot \nabla \mathbf{v}, \mathbf{w})| \le C\|\mathbf{u}\|_{\frac{1}{2}} |\mathbf{v}|_1 |\mathbf{w}|_1.$$

The case of the skew-symmetrized form follows by applying the inequalities twice, since

$$b_2(\mathbf{u}, \mathbf{v}, \mathbf{w}) = \frac{1}{2}(b_1(\mathbf{u}, \mathbf{v}, \mathbf{w}) - b_1(\mathbf{u}, \mathbf{w}, \mathbf{v})).$$

The last three inequalities (g), (h), (i) for the convective form can be derived in the following way:

$$\begin{aligned} |(\mathbf{u} \cdot \nabla \mathbf{v}, \mathbf{w})| &\le C\|\mathbf{u}\|_0 |\mathbf{v}|_1 \|\mathbf{w}\|_{0,\infty} \le C\|\mathbf{u}\|_0 |\mathbf{v}|_1 \|\mathbf{w}\|_2, \\ |(\mathbf{u} \cdot \nabla \mathbf{v}, \mathbf{w})| &\le |(\nabla \cdot \mathbf{u}, \mathbf{v} \cdot \mathbf{w})| + |(\mathbf{u} \cdot \nabla \mathbf{w}, \mathbf{v})| \\ &\le C|\mathbf{u}|_1 \|\mathbf{v}\|_0 \|\mathbf{w}\|_{0,\infty} + C\|\mathbf{u}\|_{0,6} |\mathbf{w}|_{1,3} \|\mathbf{v}\|_0 \\ &\le C|\mathbf{u}|_1 \|\mathbf{v}\|_0 \|\mathbf{w}\|_2 \end{aligned}$$

and

$$|(\mathbf{u} \cdot \nabla \mathbf{v}, \mathbf{w}) \le C\|\mathbf{u}\|_{0,6} |\mathbf{v}|_{1,3} \|\mathbf{w}\|_0 \le C|\mathbf{u}|_1 \|\mathbf{v}\|_2 \|\mathbf{w}\|_0.$$

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Finally, for the skew-symmetrized form, we have

$$\begin{aligned}|b_2(\mathbf{u},\mathbf{v},\mathbf{w})| &\le C(\|\mathbf{u}\|_0|\mathbf{v}|_1\|\mathbf{w}\|_2 + \|\mathbf{u}\|_0|\mathbf{w}|_{1,3}\|\mathbf{v}\|_{0,6})\\ &\le C\|\mathbf{u}\|_0|\mathbf{v}|_1\|\mathbf{w}\|_2,\end{aligned}$$

applying (h) and (i) for $b = b_1$ on the right-hand side of

$$b_2(\mathbf{u},\mathbf{v},\mathbf{w}) = \frac{1}{2}b_1(\mathbf{u},\mathbf{v},\mathbf{w}) - \frac{1}{2}b_1(\mathbf{u},\mathbf{w},\mathbf{v}).$$

Since the conservation form of $b = b_3(\cdot,\cdot,\cdot)$ is one additive term in the skew-symmetrized form $b = b_2$, the methods used to bound b_2 apply directly to b_3 as well.

In order to show that the correction Step III does improve the scaling properties needed for the graph norm optimality, we introduce the Galerkin projection $(Q,R) : \mathbf{Y} \to \mathbf{Y}^H$ with respect to the dual linearized problem as follows:

$$B^H[(\mathbf{w},r);(\mathbf{v}-Q(\mathbf{v},q),q-R(\mathbf{v},q))] = 0, \qquad \text{for all } (\mathbf{w},r)\in\mathbf{Y}^H, (\mathbf{v},q)\in\mathbf{Y}. \tag{2.13}$$

Note that, due to (2.11), which is equivalent to

$$\inf_{(\mathbf{v},q)\in\mathbf{Y}^H}\ \sup_{(\mathbf{u},p)\in\mathbf{Y}^H} \frac{B^H[(\mathbf{u},p);(\mathbf{v},q)]}{|||(\mathbf{u},p)|||\ |||(\mathbf{v},q)|||} \ge \gamma > 0, \tag{2.14}$$

(Q,R) is well defined. To derive an L^2-error estimate for $\mathbf{v}$-$Q(\mathbf{v},q)$, assume the $W^{2,2}$-regularity of the linearized Navier–Stokes equations. To formulate this assumption, let $(\phi,\lambda)\in\mathbf{Y}$ be the solution of the linearized Navier–Stokes equations.

Find $(\boldsymbol{\phi},\lambda)\in\mathbf{Y}$ such that, for all $(\mathbf{w},r)\in\mathbf{Y}$,

$$a(\boldsymbol{\phi},\mathbf{w}) + b(\mathbf{u},\boldsymbol{\phi},\mathbf{w}) + b(\boldsymbol{\phi},\mathbf{u},\mathbf{w}) - (p,\nabla\cdot\mathbf{w}) + (r,\nabla\cdot\boldsymbol{\phi}) = (\mathbf{f},\mathbf{w}). \tag{2.15}$$

Since $\mathbf{u}$ is a nonsingular solution, $(\boldsymbol{\phi},\lambda)$ exists uniquely. The assumption of $W^{2,2}(\Omega)$-regularity means that, for any $\mathbf{f}\in(L^2(\Omega))^d$, the solution $(\boldsymbol{\phi},\lambda)$ to (2.15) belongs to

$$(\overset{\circ}{W}{}^{1,2}(\Omega)\cap W^{2,2}(\Omega))^d \times (L_0^2(\Omega)\cap W^{1,2}(\Omega))$$

and the inequality

$$\|\boldsymbol{\phi}\|_2 + \|\lambda\|_1 \le C\|\mathbf{f}\|_0$$

holds.

LEMMA 2.3. *Let* $\mathbf{u}$ *be a nonsingular solution of the Navier–Stokes equations and suppose* $\mathbf{u}^H$ *is close enough to* $\mathbf{u}$. *Further, let the linearized Navier–Stokes equation* (2.15) *be* $W^{2,2}$ *regular. Then the projection* (Q,R) *satisfies, for* $0\le\theta\le 1$,

$$|||(\mathbf{v}-Q(\mathbf{v},q),q-R(\mathbf{v},q))||| \le C|||(\mathbf{v},q)||| \text{ for all } (\mathbf{v},q)\in\mathbf{Y}, \tag{2.16}$$

$$||\mathbf{v}-Q(\mathbf{v},q)||_\theta \le CH^{1-\theta}|||(\mathbf{v},q)||| \text{ for all } (\mathbf{v},q)\in\mathbf{Y}, \tag{2.17}$$

$$||\mathbf{v}-Q(\mathbf{v},q)||_0 + H|\mathbf{v}-Q(\mathbf{v},q)|_1 \le CH^2(||\mathbf{v}||_2 + ||q||_1), \tag{2.18}$$
$$\text{for all } (\mathbf{v},q)\in\mathbf{Y}\cap(W^{2,2}(\Omega)^d\times W^{1,2}(\Omega)).$$

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Proof. The stability of the projection follows simply by using (2.14) and the continuity of B^H, namely,

$$\begin{aligned}|||(Q(\mathbf{v},q),R(\mathbf{v},q))||| &\le \frac{1}{\gamma}\sup(\mathbf{w},r)\in\mathbf{Y}^H\frac{B^H[(\mathbf{w},r);(Q(\mathbf{v},q),R(\mathbf{v},q))]}{|||(\mathbf{w},r)|||}\\ &= \frac{1}{\gamma}\sup(\mathbf{w},r)\in\mathbf{Y}^H\frac{B^H[(\mathbf{w},r);(\mathbf{v},q)]}{|||(\mathbf{w},r)|||}\\ &\le \frac{C}{\gamma}|||(\mathbf{v},q)|||,\ \text{for all } (\mathbf{v},q)\in\mathbf{Y}.\end{aligned}$$

Now the triangle inequality gives the first assertion of the lemma. To prove the second, we denote by $(\boldsymbol{\phi}_\mathbf{f},\lambda_\mathbf{f})$ the solution of (2.15) for the special right-hand side $\mathbf{f}=\mathbf{v}-Q(\mathbf{v},q)\in(L^2(\Omega))^d$. Then a simple calculation shows that, for all $(\mathbf{w},r)\in\mathbf{Y}$,

$$B^H[(\boldsymbol{\phi}_\mathbf{f},\lambda_\mathbf{f});(\mathbf{w},r)] = (\mathbf{f},\mathbf{w}) + b(\mathbf{u}^H-\mathbf{u},\boldsymbol{\phi}_\mathbf{f},\mathbf{w}) + b(\boldsymbol{\phi}_\mathbf{f},\mathbf{u}^H-\mathbf{u},\mathbf{w}).$$

Now, setting $\mathbf{w}:=\mathbf{v}-Q(\mathbf{v},q), r:=q-R(\mathbf{v},q)$ and using the projection property, we have, for arbitrary $(\boldsymbol{\phi}^H,\lambda^H)\in\mathbf{Y}^H$,

$$\begin{aligned}||\mathbf{v}-Q(\mathbf{v},q)||_0^2 &= B^H[(\phi_\mathbf{f}-\phi^\mathbf{H},\lambda_\mathbf{f}-\lambda^\mathbf{H});(\mathbf{v}-\mathbf{Q}(\mathbf{v},\mathbf{q}),\mathbf{q}-\mathbf{R}(\mathbf{v},\mathbf{q}))]\\ &\quad + b(\mathbf{u}-\mathbf{u}^H,\phi_\mathbf{f},\mathbf{v}-\mathbf{Q}(\mathbf{v},\mathbf{q})) + \mathbf{b}(\phi_\mathbf{f},\mathbf{u}-\mathbf{u}^\mathbf{H},\mathbf{v}-\mathbf{Q}(\mathbf{v},\mathbf{q}))\\ &\le C\left\{|||(\phi_\mathbf{f}-\phi^\mathbf{H},\lambda_\mathbf{f}-\lambda^\mathbf{H})||| + |\mathbf{u}-\mathbf{u}^\mathbf{H}|_\mathbf{1}|\phi_\mathbf{f}|_\mathbf{1}\right\}\\ &\qquad\cdot|||(\mathbf{v}-Q(\mathbf{v},q),q-R(\mathbf{v},q))|||\\ &\le C(H+H^k)||\phi_\mathbf{f}||_\mathbf{2}|||(\mathbf{v},\mathbf{q})|||\\ &\le CH||\mathbf{v}-Q(\mathbf{v},q)||_0|||(\mathbf{v},q)|||.\end{aligned}$$

This inequality and the interpolation inequality

$$||\mathbf{v}-Q(\mathbf{v},q)||_\theta \le ||\mathbf{v}-Q(\mathbf{v},q)||_0^{1-\theta}|\mathbf{v}-Q(\mathbf{v},q)|_1^\theta$$

gives the result

$$||\mathbf{v}-Q(\mathbf{v},q)||_\theta \le CH^{1-\theta}|||(\mathbf{v},q)|||.$$

It remains to consider the case when $(\mathbf{v},q)$ belongs to $\mathbf{Y}\cap((W^{2,2}(\Omega))^d\times W^{1,2}(\Omega))$. Then, with the standard interpolation $(I^H\mathbf{v},J^Hp)\in\mathbf{Y}^H$, we have, again by (2.14),

$$|||(I^H\mathbf{v}-Q(\mathbf{v},q),J^hq-R(\mathbf{v},q))||| \le \frac{1}{\gamma}\sup(\mathbf{w},r)\in\mathbf{Y}^H\frac{B^H[(\mathbf{w},r);(I^h\mathbf{v}-\mathbf{v},J^hq-q)]}{|||(\mathbf{w},r)|||}.$$

Thus the triangles inequality and approximate properties of the interpolation give

$$|||(\mathbf{v}-Q(\mathbf{v},q),q-R(\mathbf{v},q))||| \le \left(1+\frac{\text{const}}{\gamma}\right)H(||\mathbf{v}||_2+||q||_1). \tag{2.19}$$

Now, repeating the arguments above and using the improved estimation (2.19) instead of (2.16), we obtain

$$||\mathbf{v}-Q(\mathbf{v},q)||_0 \le CH^2(||\mathbf{v}||_2+||q||_1). \qquad \square$$

Proof of Theorem 2.1. First, consider briefly the error of the method after Step II. The error satisfies, for all $(\mathbf{v},q)\in\mathbf{Y}^h$,

$$A^H[(\mathbf{u}-\mathbf{u}^h,p-p^h);(\mathbf{v},q)] = b(\mathbf{u}^H-\mathbf{u},\mathbf{u},\mathbf{v})$$

so that, by means of the continuity of A^H, we have, for $(I^h\mathbf{u}, J^h p) \in \mathbf{Y}^h$ and interpolant of $(\mathbf{u}, p)$ into $\mathbf{Y}^h$,

$$\begin{aligned}
|||(\mathbf{u}-\mathbf{u}^h, p-p^h)||| &\le |||(\mathbf{u}-I^h\mathbf{u}, p-J^h p)||| + |||(I^h\mathbf{u}-\mathbf{u}^h, J^h p-p^h)||| \\
&\le Ch^k(|\mathbf{u}|_{k+1}+|p|_k) + \frac{1}{\gamma}\sup(\mathbf{v},q)\in\mathbf{Y}^h \frac{A^H[(I^h\mathbf{u}-\mathbf{u}, J^h p-p);(\mathbf{v},q)]}{|||(\mathbf{v},q)|||} \\
&+ \frac{1}{\gamma}\sup(\mathbf{v},q)\in\mathbf{Y}^h \frac{b(\mathbf{u}^H-\mathbf{u},\mathbf{u},\mathbf{v})}{|||(\mathbf{v},q)|||} \\
&\le Ch^k(|\mathbf{u}|_{k+1}+|p|_k) + C||\mathbf{u}^H-\mathbf{u}||_0|\mathbf{u}|_{1+s} \\
&\le C(\mathbf{u},p)(h^k+H^{k+1}).
\end{aligned}$$

Here, as in Lemma 2.2, s is the dimension dependent parameter given by $s=\epsilon>0$ arbitrary small if $d=2$ and $s=\frac{1}{2}$ if $d=3$.

Next, we show that the error $(\mathbf{u}-\mathbf{u}^*, p-p^*)$ satisfies the following relation for all $(\mathbf{v},q)\in\mathbf{Y}^h$:

$$\begin{aligned}
B^H[(\mathbf{u}-\mathbf{u}^*, p-p^*);(\mathbf{v},q)] &= b(\mathbf{u}^H-\mathbf{u}^h, \mathbf{u}^H, \mathbf{v}-Q(\mathbf{v},q)) \\
&\quad - b(\mathbf{u}-\mathbf{u}^H, \mathbf{u}-\mathbf{u}^H, \mathbf{v}).
\end{aligned}$$

As we have already seen, $(\mathbf{u}-\mathbf{u}^h, p-p^h)$ satisfies

$$A^H[(\mathbf{u}-\mathbf{u}^h, p-p^h);(\mathbf{v},q)] = b(\mathbf{u}^H-\mathbf{u},\mathbf{u},\mathbf{v}) \qquad \text{for all } (\mathbf{v},q)\in\mathbf{Y}^h,$$

which implies

(2.20)
$$B^H[(\mathbf{u}-\mathbf{u}^h, p-p^h);(\mathbf{v},q)] = b(\mathbf{u}^H-\mathbf{u},\mathbf{u},\mathbf{v}) + b(\mathbf{u}-\mathbf{u}^h, \mathbf{u}^H, \mathbf{v}) \quad \text{for all } (\mathbf{v},q)\in\mathbf{Y}^h.$$

Step III and definition (2.13) of the projection operator (Q,R) give

$$B^H[(\mathbf{e}^H,\epsilon^H);(\mathbf{v},q)] = B^H[(\mathbf{e}^H,\epsilon^H);(Q(\mathbf{v},q),R(\mathbf{v},q))] = b(\mathbf{u}^H-\mathbf{u}^h, \mathbf{u}^H, Q(\mathbf{v},q)).$$

Consequently,

$$\begin{aligned}
B^H[(\mathbf{u}-\mathbf{u}^*, p-p^*);(\mathbf{v},q)] &= b(\mathbf{u}^H-\mathbf{u},\mathbf{u},\mathbf{v}) + b(\mathbf{u}-\mathbf{u}^h, \mathbf{u}^H, \mathbf{v}) \\
&\quad + b(\mathbf{u}^H-\mathbf{u}^h, \mathbf{u}^H, \mathbf{v}-Q(\mathbf{v},q)) - b(\mathbf{u}^H-\mathbf{u}^h, \mathbf{u}^H, \mathbf{v}) \\
&= -b(\mathbf{u}-\mathbf{u}^H, \mathbf{u}-\mathbf{u}^H, \mathbf{v}) \\
&\quad + b(\mathbf{u}^H-\mathbf{u}^h, \mathbf{u}^H, \mathbf{v}-Q(\mathbf{v},q)). \qquad \square
\end{aligned}$$

Now, the remainder of the proof of Theorem 2.1 follows by normal-type arguments applying (2.12). Indeed,

(2.21)
$$\begin{aligned}
|||(\mathbf{u}-\mathbf{u}^*, p-p^*)||| &\le |||(\mathbf{u}-I^h\mathbf{u}, p-J^h p)||| + |||(I^h\mathbf{u}-\mathbf{u}^*, J^h p-p^*)||| \\
&\le Ch^k(|\mathbf{u}|_{k+1}+|p|_k) \\
&\quad + \frac{1}{\gamma}\sup_{(\mathbf{v},q)\in Y^h} \frac{|b(\mathbf{u}^h-\mathbf{u}^H, \mathbf{u}^H, \mathbf{v}-Q(\mathbf{v},q)) - b(\mathbf{u}-\mathbf{u}^H,\ \mathbf{u}-\mathbf{u}^H, \mathbf{v})|}{|||(\mathbf{v},q)|||}.
\end{aligned}$$

Define RHS to be the following quantity:

$$\text{RHS } := b(\mathbf{u}^h-\mathbf{u}^H, \mathbf{u}^H, \mathbf{v}-Q(\mathbf{v},q)) - b(\mathbf{u}-\mathbf{u}^H, \mathbf{u}-\mathbf{u}^H, \mathbf{v}).$$

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Adding and subtracting terms gives

$$\begin{aligned}RHS &= b(\mathbf{u}^h - \mathbf{u}^H, \mathbf{u}^H - \mathbf{u}, \mathbf{v} - Q(\mathbf{v}, q)) \\ &\quad + b(\mathbf{u}^h - \mathbf{u}^H, \mathbf{u}, \mathbf{v} - Q(\mathbf{v}, q)) - b(\mathbf{u} - \mathbf{u}^H, \mathbf{u}^H, \mathbf{v}) \\ &= b(\mathbf{u}^h - \mathbf{u}, \mathbf{u}^H - \mathbf{u}, \mathbf{v} - Q(\mathbf{v}, q)) + b(\mathbf{u} - \mathbf{u}^H, \mathbf{u}^H - \mathbf{u}, \mathbf{v} - Q(\mathbf{v}, q)) \\ &\quad + b(\mathbf{u}^h - \mathbf{u}, \mathbf{u}, \mathbf{v} - Q(\mathbf{v}, q)) + b(\mathbf{u} - \mathbf{u}^H, \mathbf{u}, \mathbf{v} - Q(\mathbf{v}, q)) - b(\mathbf{u} - \mathbf{u}^H, \mathbf{u} - \mathbf{u}^H, \mathbf{v}).\end{aligned}$$

Now, $\|\mathbf{v} - Q(\mathbf{v}, q)\|_\theta \le CH^{1-\theta}|||(\mathbf{v}, q)|||, 0 \le \theta \le 1$. Thus, using the bounds of Lemma 2.2 repeatedly,

$$\begin{aligned}|b(\mathbf{u}^h - \mathbf{u}, \mathbf{u}^H - \mathbf{u}, \mathbf{v} - Q(\mathbf{v}, q))| &\le C|\mathbf{u}^h - \mathbf{u}|_1|\mathbf{u}^H - \mathbf{u}|_1|\mathbf{v} - Q(\mathbf{v}, q)|_1 \\ &\le C(h^k + H^{k+1})H^k|||(\mathbf{v}, q)|||, \\ |b(\mathbf{u} - \mathbf{u}^H, \mathbf{u}^H - \mathbf{u}, \mathbf{v} - Q(\mathbf{v}, q))| &\le C||\mathbf{u} - \mathbf{u}^H||_0^{1-s}|\mathbf{u} - \mathbf{u}^H|_1^{1+s}|\mathbf{v} - Q(\mathbf{v}, q)|_1 \\ &\le CH^{2k+1-s}|||(\mathbf{v}, q)|||, \\ |b_1(\mathbf{u}^h - \mathbf{u}, \mathbf{u}, \mathbf{v} - Q(\mathbf{v}, q))| &\le C|\mathbf{u}^h - \mathbf{u}|_1|\mathbf{u}|_{1,3}||\mathbf{v} - Q(\mathbf{v}, q)||_0 \\ &\le C(h^k + H^{k+1})H|||(\mathbf{v}, q)|||, \\ |b_1(\mathbf{u} - \mathbf{u}^H, \mathbf{u}, \mathbf{v} - Q(\mathbf{v}, q))| &\le C||\mathbf{u} - \mathbf{u}^H||_0|\mathbf{u}|_{1,\infty}||\mathbf{v} - Q(\mathbf{v}, q)||_0 \\ &\le CH^{k+2}|||(\mathbf{v}, q)|||, \\ |b(\mathbf{u} - \mathbf{u}^H, \mathbf{u} - \mathbf{u}^H, \mathbf{v})| &\le C||\mathbf{u} - \mathbf{u}^H||_0^{1-s}|\mathbf{u} - \mathbf{u}^H|_1^{1+s}|\mathbf{v}|_1 \\ &\le CH^{2k+1-s}|||(\mathbf{v}, q)|||.\end{aligned}$$

Inserting the above into (2.21) yields

$$|||(\mathbf{u} - \mathbf{u}^*, p - p^*)||| \le C(\mathbf{u}, p)(h^k + H^{k+2} + H^{2k+1-s}),$$

which is the main result of this section. ∎

REMARK 2.5. *If Algorithm* 1.1 *is posed using the skew-symmetrized trilinear form* b_2 *or the conservation form* b_3 *in lieu of the convective form* b_1*, then the following analysis predicts an error bound of* $O(h^k + H^{k+1})$ *(vs.* $O(h^k + H^{k+2})$*). The only difference to the above approach is the estimation of the following two terms:*

$$\begin{aligned}|b_2(\mathbf{u}^h - \mathbf{u}, \mathbf{u}, \mathbf{v} - Q(\mathbf{v}, q))| &\le C|\mathbf{u}^h - \mathbf{u}|_1|\mathbf{u}|_1|\mathbf{v} - Q(\mathbf{v}, q)|_1, \\ &\le C(h^k + H^{k+1})|||(\mathbf{v}, q)|||\end{aligned}$$

and

$$\begin{aligned}|b_2(\mathbf{u} - \mathbf{u}^H, \mathbf{u}, \mathbf{v} - Q(\mathbf{v}, q))| &\le C(||\mathbf{u} - \mathbf{u}^H||_0|\mathbf{u}|_{1,\infty} + |\mathbf{u} - \mathbf{u}^H|_1||\mathbf{u}||_{0,\infty})||\mathbf{v} - Q(\mathbf{v}, q)||_0, \\ &\le C(H^{k+1} + H^k)H|||(\mathbf{v}, q)|||.\end{aligned}$$

The same estimates hold if the conservation form $b = b_3$ *has been used. This predicts an error* $O(h^k + H^{k+1})$ *in the graph norm which is no better than stopping at Step* II, [19]. *Nevertheless, in the next section, we show that even in this case Step* III *improves the* L^2*-error over the Step* II *approximation.*

REMARK 2.6 (the energy norm error of Algorithm 1.2). *The approximate solution given by Algorithm* 1.2 *can be shown to satisfy the following approximate Galerkin orthogonality relations for all* $(\mathbf{v}, q) \in \mathbf{Y}^h$*:*

$$B^H[(\mathbf{u} - \mathbf{u}^h, p - p^h); (\mathbf{v}, q)] = b(\mathbf{u}^H - \mathbf{u}, \mathbf{u} - \mathbf{u}^H, \mathbf{v}) \tag{2.22}$$

$$B^H[(\mathbf{u}-\ \mathbf{u}^*, p-p^*);(\mathbf{v},q)] = b(\mathbf{u}-\mathbf{u}^h, \mathbf{u}-\mathbf{u}^h, \mathbf{v}) - b(\mathbf{u}-\mathbf{u}^H, \mathbf{u}-\mathbf{u}^h, \mathbf{v}) \\ -\ b(\mathbf{u}-\mathbf{u}^h, \mathbf{u}-\mathbf{u}^H, \mathbf{v}) + b(\mathbf{u}^h-\mathbf{u}^H, \mathbf{u}^h-\mathbf{u}^H, Q(\mathbf{v},q)-\mathbf{v}). \tag{2.23}$$

From (2.22)*, we get the estimate* (see also [16])

$$|||(\mathbf{u}-\mathbf{u}^h, p-p^h)||| \le C(h^k + H^{2k+1-s}). \tag{2.24}$$

Proceeding with the error analysis as in the proof of Theorem 2.1 *results in the following error bound for Algorithm* 1.2:

$$|||(\mathbf{u}-\mathbf{u}^*, p-p^*)||| \le C(h^k + H^{2k+1-s})$$

due to the term $b(\mathbf{u}-\mathbf{u}^H, \mathbf{u}-\mathbf{u}^H, Q(\mathbf{v},q)-\mathbf{v})$*. This error bound is no better than the graph norm error* (2.24)*, the result of stopping at Step* II *in Algorithm* 1.2.

3. The L^2-estimate. We shall now give an L^2-error estimate for Algorithm 1.1. This L^2-estimate is interesting because it establishes that the two-level method can be extended directly to a multilevel method. The L^2-error analysis is based first upon the assumptions needed to obtain the $W^{1,2}(\Omega)$ estimate of the previous section and second, the assumption that the following linearized adjoint problem (3.1) is $W^{2,2}(\Omega)$ regular (see Remark 2.1).

Find $(\phi, \lambda) \in \mathbf{Y}$ such that, for all $(\mathbf{w}, r) \in \mathbf{Y}$,

$$a(\mathbf{w},\phi) + \mathbf{b}(\mathbf{u},\mathbf{w},\phi) + \mathbf{b}(\mathbf{w},\mathbf{u},\phi) + (\lambda, \nabla\cdot\mathbf{w}) - (\mathbf{r}, \nabla\cdot\phi) = (\mathbf{g},\mathbf{w}). \tag{3.1}$$

Since $\mathbf{u}$ is a nonsingular solution, (ϕ,λ) exists uniquely. The assumption that the linearized adjoint problem is $W^{2,2}(\Omega)$ regular means that, for any $\mathbf{g} \in (L^2(\Omega))^d$, there is a solution (ϕ,λ) belonging to

$$(\overset{\circ}{W}{}^{1,2}(\Omega) \cap W^{2,2}(\Omega))^2 \times (L_0^2(\Omega) \cap W^{1,2}(\Omega))$$

and the inequality

$$||\phi||_{\mathbf{2}} + ||\lambda||_{\mathbf{1}} \le \mathbf{C}||\mathbf{g}||_{\mathbf{0}} \tag{3.2}$$

holds. This $W^{2,2}$-regularity assumption is also *implicit* in the previous section as well, in the form of the assumed L^2-optimality of the coarse mesh approximation.

THEOREM 3.1. *Suppose* $|||(\mathbf{u}-\mathbf{u}^*, p-p^*)||| \le C(\mathbf{u},p)h^k$*, where* $\mathbf{u}^*, p^*$ *are given by Algorithm* 1.1*. Suppose also that the conditions of Theorem* 2.1 *hold and that the solution of* (3.1) *satisfies* (3.2)*. Then, for the skew-symmetrized form* $b = b_2$ *and conservation form* $b = b_3$*, there holds*

$$\|\mathbf{u}-\mathbf{u}^*\|_0 \le C(\mathbf{u},p)(h^{k+1} + H^{k+2} + H^2h^k), \tag{3.3a}$$

$$\|\mathbf{u}-\mathbf{u}^*\|_0 \le C(\mathbf{u},p)h^{k+1}, \ \textit{provided } h = O(H^{1+\frac{1}{k+1}}). \tag{3.3b}$$

If the convective form $b = b_1$ *is used in Algorithm* 1.1 *instead, then*

$$\|\mathbf{u}-\mathbf{u}^*\|_0 \le C(\mathbf{u},p)(h^{k+1} + h^kH^2 + H^{\min(k+3,2k+1)} + hH^{k+1}), \tag{3.4a}$$

$$\|\mathbf{u}-\mathbf{u}^*\|_0 \le C(\mathbf{u},p)h^{k+1}, \ \textit{provided } h = O(H^{1+\min\ (\frac{1}{k},\frac{k}{k+1})}). \tag{3.4b}$$

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TABLE 3.1
Scalings $h = H^2$ for L^2-quasi-optimality for Algorithm 1.2.

Scalings	$\|\mathbf{u}-\mathbf{u}^h\|_0$	$\|\mathbf{u}-\mathbf{u}^*\|_0$
Linear elements ($k=1$)	$h = H^{3/2}$	$h = H^{7/4}$
Quadratic elements ($k=2$)	$h = H^{5/3}$	$h = H^{15/8}$
$\cdots$	$\cdots$	$\cdots$
degree k elements	$h = H^{2-\frac{1}{k+1}}$	$h = H^{2-\frac{1}{2(k+1)}}$

REMARK 3.1 (the L^2-error of Algorithm 1.2). *If the L^2-error of Algorithm* 1.2 *is analyzed by the same techniques and under the same assumptions as Theorem* 3.1, *the following error estimate results:*

$$\|\mathbf{u}-\mathbf{u}^*\|_0 \le C(\mathbf{u},p)(h^{k+1} + H^{2k+3/2} + H^{k+1}h^k).$$

As expected, this is better than the one resulting from Algorithm 1.1. *However it is only very slightly better than stopping Algorithm* 1.2 *at Step* II. *Indeed, in [*18, *Thm.* 2.2*] it is proven that*

$$\|\mathbf{u}-\mathbf{u}^h\|_0 \le C(\mathbf{u},p)(h^{k+1} + H^{2k+3/2} + H^{2k+1}).$$

Thus, for Algorithm 1.2, *the convergence rates of $\|\mathbf{u}-\mathbf{u}^h\|_0$ and $\|\mathbf{u}-\mathbf{u}^*\|_0$ are quasi-optimal ($\le C(\mathbf{u},p)h^{k+1}$) provided the scalings given in Table* 3.1 *hold.*

Thus, as Tables 1.1 *and* 3.1 *show, Step* III *of Algorithm* 1.1 *and* 1.2 *improves the L^2-error of Algorithm* 1.2 *somewhat and the L^2-error of Algorithm* 1.1 *dramatically.*

Proof of Theorem 3.1. First, the solution (ϕ,λ) of (3.1) satisfies

$$B^H[(\mathbf{w},r);(\phi,\lambda)] = \mathbf{b}(\mathbf{u^H}-\mathbf{u},\mathbf{w},\phi) + \mathbf{b}(\mathbf{w},\mathbf{u^H}-\mathbf{u},\phi) + (\mathbf{g},\mathbf{w}) \quad \text{for all } (\mathbf{w},\mathbf{r}) \in \mathbf{Y}.$$

We set $\mathbf{g} := \mathbf{u}-\mathbf{u}^* \in (L^2(\Omega))^d$ and $(\mathbf{w},r) := (\mathbf{u}-\mathbf{u}^*, p-p^*)$. This gives

$$\begin{aligned}||\mathbf{u}-\mathbf{u}^*||_0^2 &= B^H[(\mathbf{u}-\mathbf{u}^*,p-p^*);(\phi,\lambda)]\\ &\quad + b(\mathbf{u}-\mathbf{u}^H,\mathbf{u}-\mathbf{u}^*,\phi) + \mathbf{b}(\mathbf{u}-\mathbf{u}^*,\mathbf{u}-\mathbf{u^H},\phi),\end{aligned}$$

and, taking into account the error equation (2.20), we have

$$\begin{aligned}||\mathbf{u}-\mathbf{u}^*||_0^2 &= B^H[(\mathbf{u}-\mathbf{u}^*,p-p^*);(\phi-\phi^{\mathbf{h}},\lambda-\lambda^{\mathbf{h}})]\\ &\quad + b(\mathbf{u}-\mathbf{u}^H,\mathbf{u}-\mathbf{u}^*,\phi) + \mathbf{b}(\mathbf{u}-\mathbf{u}^*,\mathbf{u}-\mathbf{u^H},\phi)\\ &\quad + b(\mathbf{u}-\mathbf{u}^H,\mathbf{u}-\mathbf{u}^H,\phi^{\mathbf{h}})\\ &\quad - b(\mathbf{u}^H-\mathbf{u}^h,\mathbf{u}^H,\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}})).\end{aligned} \tag{3.5}$$

We estimate the right-hand side term-by-term:

$$\begin{aligned}B^H[(\mathbf{u}-\mathbf{u}^*,p-p^*);&(\phi-\phi^{\mathbf{h}},\lambda-\lambda^{\mathbf{h}})]\\ &\le |||(\mathbf{u}-\mathbf{u}^*,p-p^*)|||\ |||(\phi-\phi^{\mathbf{h}},\lambda-\lambda^{\mathbf{h}})|||\\ &\le Ch^{k+1}(||\phi||_{\mathbf{2}} + ||\lambda||_{\mathbf{1}})\\ &\le Ch^{k+1}||\mathbf{u}-\mathbf{u}^*||_0,\end{aligned}$$

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$$
\begin{aligned}
|b(\mathbf{u}-\mathbf{u}^H,\mathbf{u}-\mathbf{u}^*,\phi)| &\le C||\mathbf{u}-\mathbf{u}^H||_0|\mathbf{u}-\mathbf{u}^*|_1||\phi||_{\mathbf{2}}\\
&\le CH^{k+1}h^k||\mathbf{u}-\mathbf{u}^*||_0,\\
|b(\mathbf{u}-\mathbf{u}^*,\mathbf{u}-\mathbf{u}^H,\phi)| &\le C|\mathbf{u}-\mathbf{u}^*|_1||\mathbf{u}-\mathbf{u}^H||_0||\phi||_{\mathbf{2}}\\
&\le Ch^kH^{k+1}||\mathbf{u}-\mathbf{u}^*||_0,\\
|b(\mathbf{u}-\mathbf{u}^H,\mathbf{u}-\mathbf{u}^H,\phi^{\mathbf{h}})| &\le |b(\mathbf{u}-\mathbf{u}^H,\mathbf{u}-\mathbf{u}^H,\phi^{\mathbf{h}}-\phi)|+|\mathbf{b}(\mathbf{u}-\mathbf{u}^{\mathbf{H}},\mathbf{u}-\mathbf{u}^{\mathbf{H}},\phi)|\\
&\le C||\mathbf{u}-\mathbf{u}^H||_0^{1-s}|\mathbf{u}-\mathbf{u}^H|_1^{1+s}|\phi^{\mathbf{h}}-\phi|_{\mathbf{1}}+\mathbf{C}||\mathbf{u}-\mathbf{u}^{\mathbf{H}}||_{\mathbf{0}}|\mathbf{u}-\mathbf{u}^{\mathbf{H}}|_{\mathbf{1}}||\phi||_{\mathbf{2}}\\
&\le C(H^{2k+1-s}h+H^{2k+1})||\mathbf{u}-\mathbf{u}^*||_0.
\end{aligned}
$$

To estimate the last term $R1 := b(\mathbf{u}^H-\mathbf{u}^h,\mathbf{u}^H,\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}}))$, we use the splitting

$$
\begin{aligned}
R1 &= b(\mathbf{u}^H-\mathbf{u},\mathbf{u}^H-\mathbf{u},\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}}))+\mathbf{b}(\mathbf{u}^{\mathbf{H}}-\mathbf{u},\mathbf{u},\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}}))\\
&\quad+b(\mathbf{u}-\mathbf{u}^h,\mathbf{u}^H-\mathbf{u},\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}}))+\mathbf{b}(\mathbf{u}-\mathbf{u}^{\mathbf{h}},\mathbf{u},\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}})).
\end{aligned}
\tag{3.6}
$$

Now, due to Lemma 2.3 and $\phi\in(\mathbf{W^{2,2}}(\Omega))^{\mathbf{d}}$, we have

$$
\begin{aligned}
|\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}})|_{\mathbf{1}} &\le |\phi^{\mathbf{h}}-\phi|_{\mathbf{1}}+|\phi-\mathbf{Q}(\phi,\lambda)|_{\mathbf{1}}+|\mathbf{Q}(\phi-\phi^{\mathbf{h}},\lambda-\lambda^{\mathbf{h}})|_{\mathbf{1}}\\
&\le Ch||\phi||_{\mathbf{2}}+\mathbf{CH}||\phi||_{\mathbf{2}}+\mathbf{C}|||(\phi-\phi^{\mathbf{h}},\lambda-\lambda^{\mathbf{h}})|||\\
&\le CH||\mathbf{u}-\mathbf{u}^*||_0
\end{aligned}
$$

and

$$
\begin{aligned}
||\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}})||_{\mathbf{0}} &\le Ch^2||\phi||_{\mathbf{2}}+\mathbf{CH^2}||\phi||_{\mathbf{2}}+\mathbf{C}\ |||(\phi-\phi^{\mathbf{h}},\lambda-\lambda^{\mathbf{h}})|||\\
&\le C(H^2+h)||\mathbf{u}-\mathbf{u}^*||_0.
\end{aligned}
$$

The first, third, and fourth terms in (3.6) are bounded according to

$$
\begin{aligned}
|b(\mathbf{u}^H-\mathbf{u},\mathbf{u}^H-\mathbf{u},\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}}))| &\le C|\mathbf{u}^H-\mathbf{u}|_1^2\ |\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}})|_{\mathbf{1}}\\
&\le CH^{2k+1}||\mathbf{u}-\mathbf{u}^*||_0,\\
|b(\mathbf{u}-\mathbf{u}^h,\mathbf{u}^H-\mathbf{u},\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}}))| &\le C|\mathbf{u}-\mathbf{u}^h|_1|\mathbf{u}^H-\mathbf{u}|_1|\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}})|_{\mathbf{1}}\\
&\le Ch^kH^{k+1}||\mathbf{u}-\mathbf{u}^*||_0,\\
|b(\mathbf{u}-\mathbf{u}^h,\mathbf{u},\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}}))| &\le C|\mathbf{u}-\mathbf{u}^h|_1||\mathbf{u}||_2||\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}})||_{\mathbf{0}}\\
&\le Ch^k(H^2+h)||\mathbf{u}-\mathbf{u}^*||_0.
\end{aligned}
$$

The estimation of the second term in (3.6) is different for the cases $b=b_1$ and $b=b_2$ or $b=b_3$, respectively. We have

$$
\begin{aligned}
|b_1(\mathbf{u}^H-\mathbf{u},\mathbf{u},\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}}))| &\le C||\mathbf{u}^H-\mathbf{u}||_0|\mathbf{u}|_{1,\infty}||\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}})||_{\mathbf{0}}\\
&\le CH^{k+1}(H^2+h)||\mathbf{u}-\mathbf{u}^*||_0
\end{aligned}
$$

and

$$
\begin{aligned}
|b_2(\mathbf{u}^H-\mathbf{u},\mathbf{u},\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}}))| &\le C\big\{H^{k+1}(H^2+h)||\mathbf{u}-\mathbf{u}^*||_0\\
&\quad+||\mathbf{u}^H-\mathbf{u}||_0|\phi^{\mathbf{h}}-\mathbf{Q}(\phi^{\mathbf{h}},\lambda^{\mathbf{h}})|_{\mathbf{1}}||\mathbf{u}||_{\mathbf{2}}\big\}\\
&\le C\big(H^{k+2}+hH^{k+1}\big)||\mathbf{u}-\mathbf{u}^*||_0.
\end{aligned}
$$

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Putting all estimates of the right-hand side of (3.5) together, we have, for the convective form $b = b_1$,

$$||\mathbf{u} - \mathbf{u}^*||_0 \leq C\big(h^{k+1} + h^k H^2 + H^{\min(k+3,2k+1)} + H^{k+1}h\big),$$

whereas the skew-symmetrized form $b = b_2$ and the conservation form $b = b_3$ give only

$$||\mathbf{u} - \mathbf{u}^*||_0 \leq C\big(h^{k+1} + h^k H^2 + H^{k+2}h\big). \qquad \square$$

Acknowledgments. We thank Professors Steve McCormick and Patrick Rabier and an unknown referee for their suggestions, questions, and comments which have improved this paper.

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